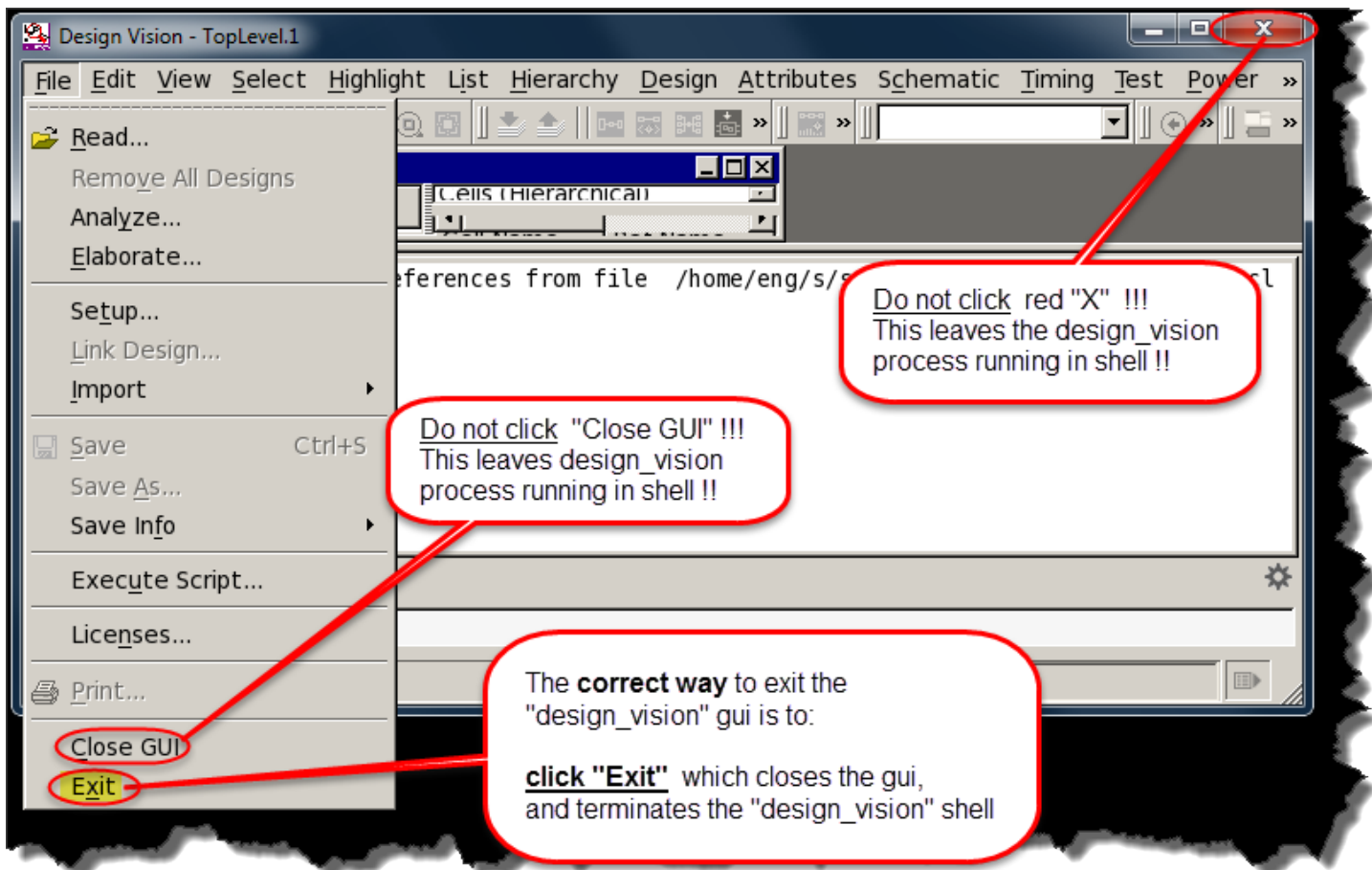


Design Vision by Synopsys

Properly quit Design Vision

We have limited number of Synopsys licenses. If you fail running Design Vision and reported an error stating all of the licenses are occupied, it means no available license.

To quit Design Vision, DO NOT simply click on the close button on the top right corner. Instead, you should go to File -> Exit to properly quit Design Vision and release occupied license.



1. Synthesis

Generate library.db

Before Design Synthesis, you need to run the Library compiler first to convert the .lib file provided to .db format.

Start Design Vision

Go to your Synopsys working directory first.

```
cd ~/cad/synopsys
```

```
~/cad/synopsys} ls  
command.log  default.svf  library.db  library.lib
```

Please source the below profile whenever you are using Synopsys tools

```
. /proj/cad/startup/profile.synopsys_2018
```

To run Design Vision , type the following to start Design Vision:

```
design_vision &
```

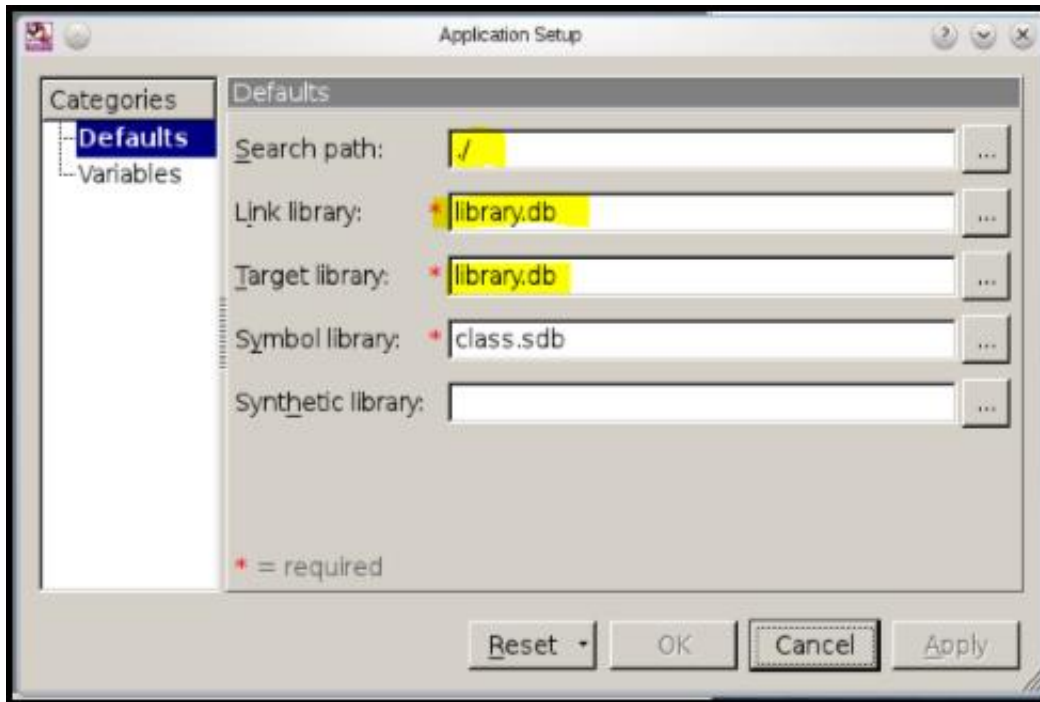
Setup design library by selecting File->Setup. Then click OK.



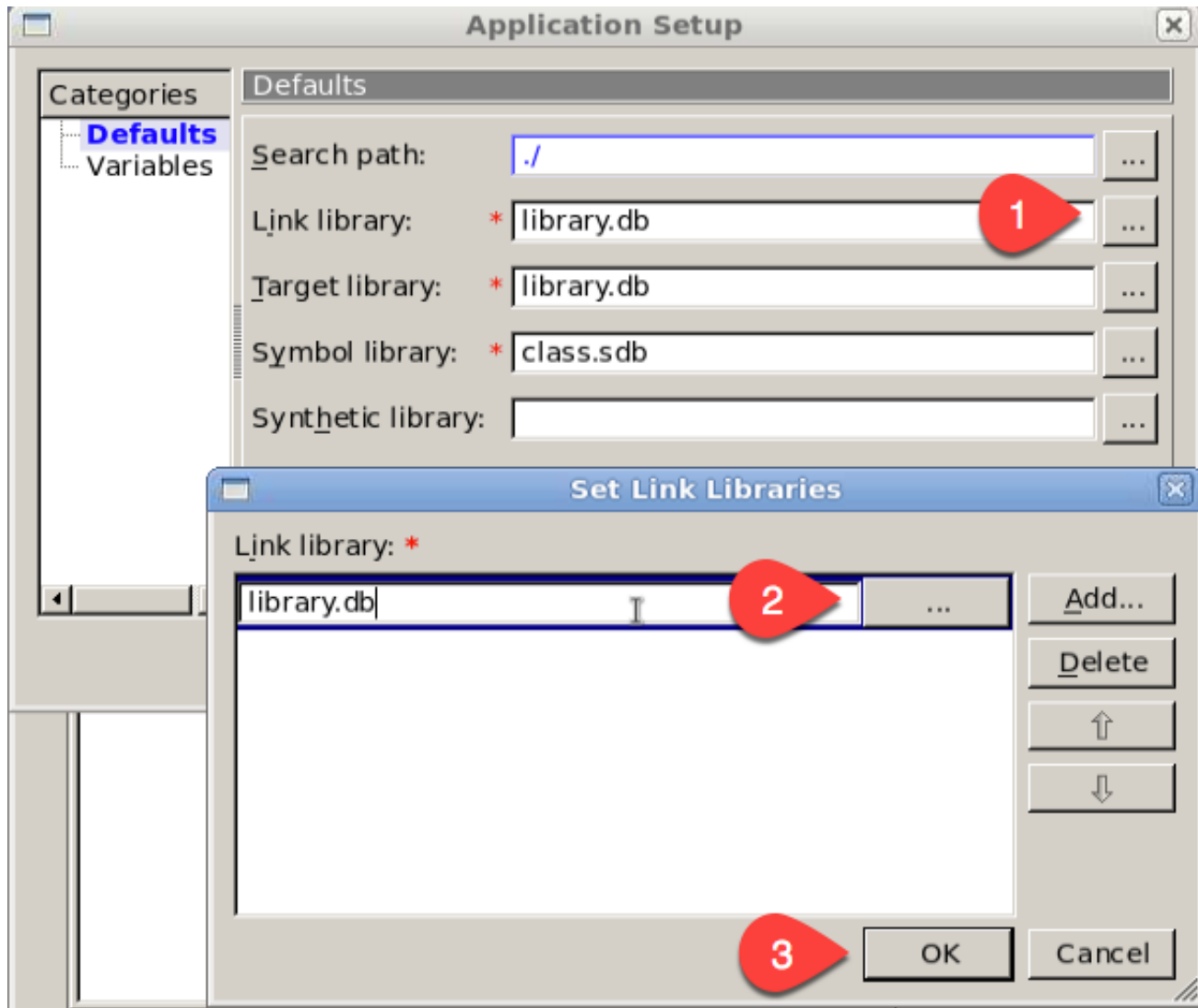
Note

The library.db is the library you generate using the Library compiler in the previous step.

- Make sure the search path is current directory i.e. "./"
- The name of link library and target library need to be "library.db"



If you store `library.db` at other location, you can not use default setting, set library location as following.



Read Design

Verilog Design Entry

Move your verilog files (not testbench) from Project 1 to your synopsys folder.

Open your design by selecting **File->Read** from the menu bar. In the **Read File** dialog box, select your verilog file and click **OK**. Ctrl+click your verilog files in order if you have multiple.

VHDL Design Entry

File -> Analyze and then, click **Add**, and add your file.

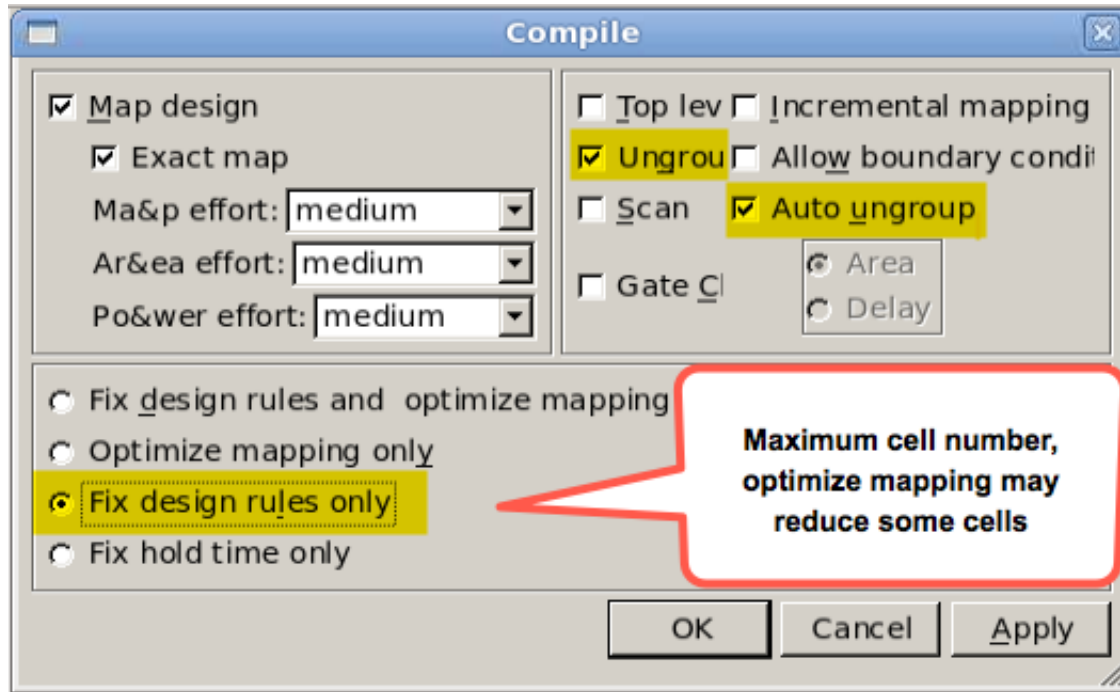
File -> Elaborate and then, click **OK**.

Note that you have just read in your design, You have not compiled or mapped it into digital gates yet. You need to do that next.

Compile Design

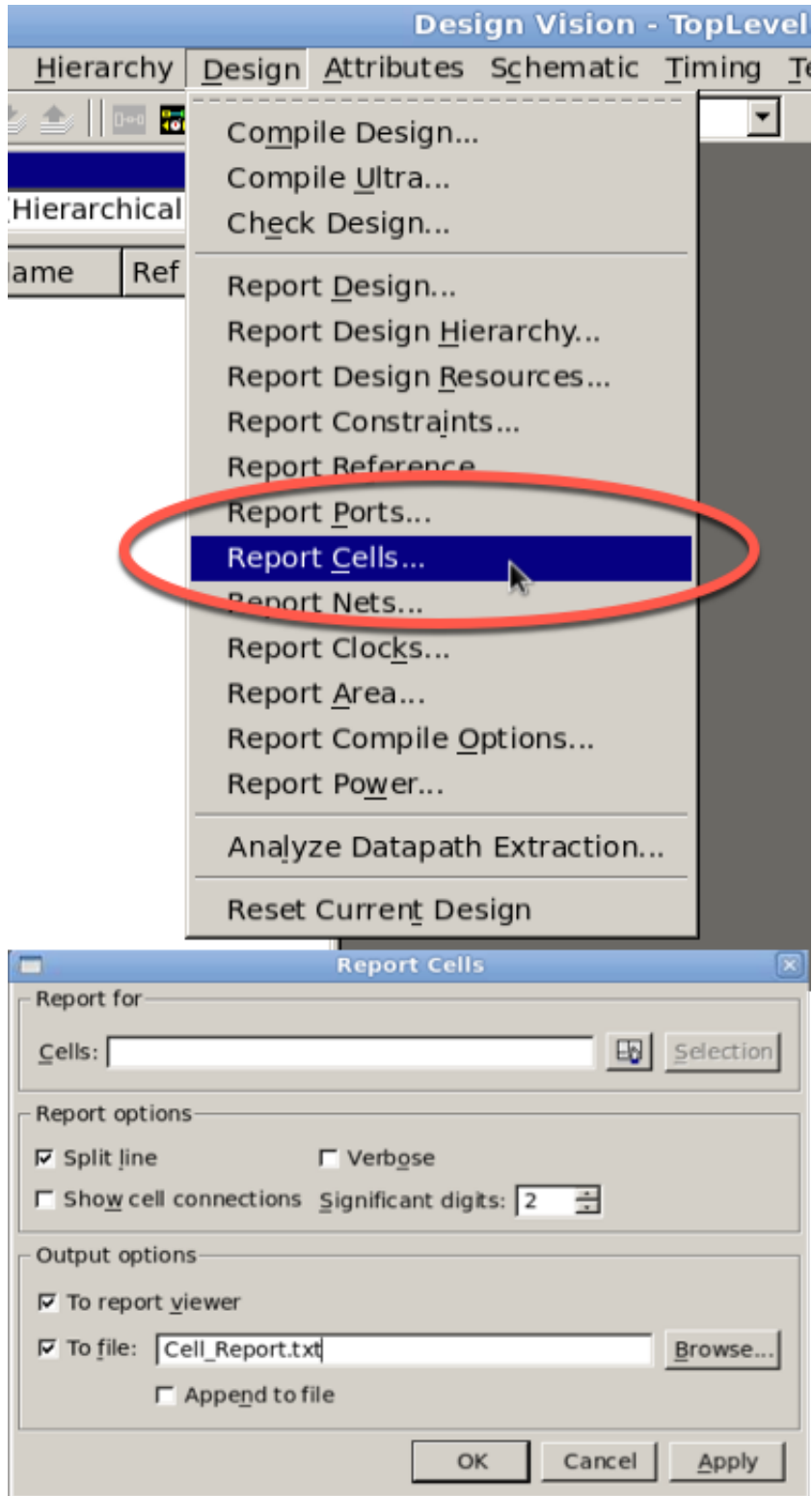
To do that select **Design->Compile Design** from the menu bar and click OK in the window that pops up.

You can check options **Ungroup all** and **Auto ungroup** for best results to flatten your verilog netlist. Then click on **Ok**



Cell Report

To check your cell number, select **Design -> Report Cells** and then **OK**. You will see the list of cells and total number of them.



Sample Cell Report

Cell	Type	Library	Value
U46	inv	library	1.000000
U47	oai22	library	2.000000
U48	inv	library	1.000000
U49	nand2	library	1.000000
U50	oai22	library	2.000000
U51	inv	library	1.000000
U52	nor2	library	1.000000
U53	inv	library	1.000000
U54	oai22	library	2.000000
U55	nor2	library	1.000000
U56	inv	library	1.000000
U57	inv	library	1.000000
U58	oai22	library	2.000000
U59	nor2	library	1.000000
U60	inv	library	1.000000
U61	inv	library	1.000000
U62	oai22	library	2.000000
U63	nor2	library	1.000000
U64	inv	library	1.000000
U65	inv	library	1.000000
U66	oai22	library	2.000000
U67	nor2	library	1.000000
U68	inv	library	1.000000
U69	inv	library	1.000000
U70	nand2	library	1.000000
U71	inv	library	1.000000
Total 69 cells			126.000000
**** End Of Report ****			

Make sure all the instance are mapped from the library.

If there are some instances like ****SEQGEN**** or ****FFGEN**** from the GTECH library then your mapping is incorrect.

Always @ (signal, ...) will infer latches and should be avoided !

Since we will design a positive/negative edge triggered Flip flop, and no latch. Your mapped netlist can only use Flip flops, instead of latches. So your code must not infer latches and should be written in a synchronous fashion.

About Group/Ungroup

The **group** command groups cells (instances) or related components in the design into a new module, creating a new level of hierarchy.

The grouped cells are replaced by a new instance (cell) that references the new module.

Removing a level of hierarchy is called **ungrouping**. Ungrouping removes (or collapses) the level of hierarchy of the identified subdesign and merges the subdesign with the surrounding logic.

If we choose to ungroup, Design Vision will take all of the logic within the module and combine it with the logic at other levels of the design. This can obtain area benefits by possibly combining redundant logic.

Example:

out_reg[7]	dff	library	out_reg[0]	dff	library
out_reg[8]	dff	library	out_reg[1]	dff	library
out_reg[9]	dff	library	out_reg[2]	dff	library
out_reg[10]	dff	library	out_reg[3]	dff	library
out_reg[11]	dff	library	out_reg[4]	dff	library
out_reg[12]	dff	library	out_reg[5]	dff	library
out_reg[13]	dff	library	out_reg[6]	dff	library
out_reg[14]	dff	library	out_reg[7]	dff	library
out_reg[15]	dff	library	out_reg[8]	dff	library
r48	alu_dw02_mult_0		out_reg[9]	dff	library
			out_reg[10]	dff	library
r179	alu_dw01_addsub_0		out_reg[11]	dff	library
			out_reg[12]	dff	library
r181	alu_dw01_addsub_1		out_reg[13]	dff	library
			out_reg[14]	dff	library
r184	alu_dw01_add_2		out_reg[15]	dff	library
-----			-----		
Total 419 cells			Total 1997 cells		
			Removing design 'alu'		
***** End Of Report *****			***** End Of Report *****		

The above picture shows the cell reports from a compile result without checking the **ungroup** option (left) v.s. the one with **ungroup** option (right).

The left result has some cells that are not from the library we use, which is not allowed. While the cells in right result are all from our library.

That's why you get more cell count with **ungroup**.

So my suggestion is: If you compile without ungroup and get the cells required in the project description AND all the cells are from the library, that's fine. Otherwise, compile with **ungroup** option.

Create Mapped Netlist

To create a verilog (or VHDL) mapped netlist for your design, select **File->Save As** again.

Specify the filename as your original verilog (or VHDL) file with the addition of `_syn.v` (i.e.: original: fsm.v ... output: fsm_syn.v) and select `verilog (or VHDL)` as the file format from the drop-down menu. Click OK.

Now close Synopsys Design Vision by selecting **File->Exit**.

2. Prepare Mapped Netlist

The resulting verilog (or VHDL) file is a gate-level netlist of your design.

It describes the circuit as a network of devices with names and pins as specified by the Synopsys library.

The netlist is not yet complete, however, as we need to add module descriptions for each of the devices as well.

If you used the Synopsys library file without making any changes, you should be able to use the following file without modification.

Otherwise, you may need to apply changes as you did with the library file.

The following file is a header that can be attached to the verilog (or VHDL) netlist directly, and contains module descriptions for the logic gates that Synopsys constructed the netlist with.

Copy the file to your Synopsys directory by typing:

For Verilog

```
cp /home/cad/startup/EE6325/aux/header.v .
```

For VHDL

```
cp /home/cad/startup/EE6325/aux/header.vhdl .
```

Adding this to the verilog (or VHDL) mapped netlist, concatenate the two files (header.v first) together or edit the netlist and copy-paste the header into the beginning.

Your mapped verilog (or VHDL) netlist is now ready for verification. You should be able to reattach your test module with minor modifications and verify the functionality of the netlist as compared to the behavioral-level design.

Remember to remove the test module when importing your verilog (or VHDL) netlist into Cadence or Silicon Ensemble later on.

The final file(s) = (header + mapped netlist) + original testbench

NOTICE for VHDL users

1. I didn't test VHDL header. If there is an error please let me know. Thanks!
2. ENCOUNTER ONLY TAKES VERILOG NETLIST. So to use Encounter, you have to import VHDL file and then select VERILOG type for the output mapped netlist in Design Vision!!

3. Simulate Mapped Netlist

You should be able to simulate your mapped (structural) verilog code using the same testbench circuit used to simulate your behavioral code. You can simulate it with [ModelSim](#), [Verilog-XL](#), etc.

when simulating your mapped code it is necessary to initialize the values of all inputs at time=0.

Compare the simulation waveforms before and after synthesis, they should exactly match.

Kill Design Vision

Again, please quit Design Vision properly by File -> Exit and release occupied license. However, if you have mistakenly quit it, you can use following command to kill `design_vision` process.

```
pkill -u `id | cut -c 5-10` -f design_vision
```

Last update: September 1, 2022 13:15:33